Photonic Optical Neural Network Training Processor

Reciprocal-Transmittance Backpropagation for an Incoherent Intensity-Domain Photonic Processor

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A sequel to "Photonic Optical Neural Network Inference Processor" [1]

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Abstract

Optical accelerators that compute a neuron's pre-activation as $\sum_i x_i w_{ij}$ by attenuating N mutually incoherent beams and summing their power at a photodetector are attractive for embedded inference because they need no interferometric stability: addition is a direct consequence of incoherent intensity superposition, and multiplication is a direct consequence of passive attenuation. This simplicity is usually purchased at the price of training: the dominant photonic training demonstrations to date rely on coherent, phase-stable circuits (interferometer meshes) whose internal reciprocity supplies the gradient, a route unavailable once phase information is discarded at every detector by design. This paper asks how far training can be pushed for the *incoherent* class of device specifically, using only the two primitives — attenuation and incoherent superposition — that already justify its simplicity. We show that a strictly weaker property than coherent reciprocity, namely the direction-independence of a passive attenuator's transmittance, is already sufficient to obtain the matrix-transpose product backpropagation requires, via a second optical relay sharing the same physical attenuating plane. We extend a differential push–pull encoding to carry signed error signals through this second relay, show that the activation-derivative term needed between layers is already latent in the forward measurement, and confront honestly the one operation this approach cannot supply: the weight-update outer product, which a simple counting argument shows cannot be recovered from aggregate row/column detectors, and which a literal photosensitive-write scheme fails to supply for incoherent light by a dose-additivity argument. Two resolutions are compared — a conservative hybrid scheme with a local electronic multiply-accumulate write, proposed as the primary result, and a more speculative two-timescale scheme exploiting genuine coherent holographic recording during a rare, slow consolidation phase, offered only as an open perspective. A closed-form latency model is derived, properly accounting for the settling time of the attenuating plane itself — the same dominant timescale PONNIP's own analysis identified for inference — showing that training is genuinely slower than inference unless updates are batched before being written, and the paper closes with an explicit, unresolved list of open problems.

Contents

1 Motivation

This paper is a direct sequel to the Photonic Optical Neural Network Inference Processor (PONNIP) [1], which derived, from first principles, an incoherent intensity-domain architecture for the *inference* side of a photonic accelerator: attenuation realises scalar multiplication, intensity superposition realises addition, and rectification at the photodetector supplies the network nonlinearity at no additional cost, all without requiring any interferometric stability. PONNIP deliberately left training out of scope, loading weights once at deployment time from a conventionally, off-line trained model. This is common practice for a sequel paper in this area — citing one's own prior architecture paper and stating explicitly which open question it deferred — and is exactly what is done here: the question taken up in this paper is whether the same two incoherent primitives that make PONNIP's inference path simple can also supply, on-chip, the gradient information needed to train it.

Photonic neural-network accelerators based on coherent interferometer meshes [2] can compute a matrix-vector product as a unitary linear optical transformation, and this same coherence and reciprocity has been exploited directly to train such meshes in situ, by launching an adjoint optical field backward through the device and reading local power asymmetries at each phase shifter [5, 6]. PONNIP's structurally different, incoherent class of device instead represents inputs and weights as intensities and transmittances of *mutually incoherent* sources [3, 4], discarding phase information at every detector by design — and with it, the obvious route to an adjoint-field training method.

This raises a specific question, distinct from "can photonic neural networks be trained optically" (already answered affirmatively for the coherent case [5, 6, 7] and side-stepped for any physical hardware class by training a digital twin instead [8]): *can the incoherent, attenuation-superposition class of device be trained using only the same two primitives, attenuation and incoherent superposition, that justify its existence in the first place?* The remainder of this paper answers that question as far as a first-principles derivation allows, and states plainly where the answer runs out.

2 The Forward Primitives and the Backpropagation Requirement

Two physical facts realise a multiply-accumulate operation incoherently. First, for N mutually incoherent optical sources of intensity $x_1, \ldots, x_N$ incident on a common photodetector, the cross terms in $|E_1 + \cdots + E_N|^2$ time-average to zero because the relative phases are uncorrelated, leaving a photocurrent $i \propto \sum_i x_i$: incoherent intensities add. Second, a beam of intensity x_i passing through a passive attenuating element of transmittance $w_{ij} \in [0, 1]$ emerges with intensity $x_i w_{ij}$, by definition of transmittance. Arranging N inputs as columns and M output neurons as rows of a two-dimensional transmittance array, illuminating column i with x_i and collecting row j on a dedicated detector D_j gives

$$z_j = \sum_{i=1}^{N} x_i w_{ij}, \tag{1}$$

the pre-activation of neuron j. Because optical power is non-negative, an electronic readout that does not synthetically restore a negative offset reports exactly $\mathrm{ReLU}(z_j) = \max(z_j, 0)$ at no additional component cost — the rectifying nonlinearity between layers is a property of intensity detection, not an added operation. Signed weights are realised by a differential pair $w_{ij} = w_{ij}^+ - w_{ij}^-$, each component in $[0, 1]$, with the true pre-activation recovered by an electronic subtraction of two detector banks, $o_j^+ - o_j^-$. None of this is new; it is the standard toolkit of incoherent, broadcast-and-weight-style photonic processors [3, 4], restated here only to fix notation.

Training by gradient descent additionally requires, at every layer l with weights $W^{(l)}$,

$$\delta_j^{(l)} = \Big(\textstyle\sum_k \delta_k^{(l+1)} \, w_{jk}^{(l)} \Big) \nVdash \Big[z_j^{(l)} > 0 \Big] \,, \qquad \Delta w_{ij}^{(l)} = -\eta \, x_i^{(l)} \, \delta_j^{(l)}. \tag{2}$$

Three quantities not supplied by Eq. (1) are needed: the matrix-transpose product inside the parentheses, the binary gate $\nVdash[z_j > 0]$, and the outer product $x_i \delta_j$. Sections 3–6 address each.

3 The Matrix-Transpose Product via Reciprocal Transmittance

3.1 A weaker substitute for coherent reciprocity

A passive amplitude attenuator's transmittance does not depend on the direction light travels through it: a thin absorptive film obeying $T = e^{-\alpha \ell}$, or a liquid-crystal cell operated in amplitude mode, attenuates a beam by the same factor w_{ij} whether the beam travels from the "input" side to the "output" side or the reverse. This is close to definitional for a passive, non-gyrotropic medium and is a markedly weaker property than the coherent phase reciprocity exploited by adjoint in-situ training of interferometer meshes [5, 6], which requires measuring local interference at internal nodes under a phase-stable adjoint excitation. The claim here needs no phase information at all — only that an *intensity* attenuation factor is symmetric in direction, which holds even though every detector in the forward path has already destroyed phase by design.

3.2 A second relay sharing the same plane

The forward path of Section 2 uses one cylindrical-optics relay, focusing along the column axis so that all light leaving row j converges on detector D_j regardless of which column it originated from, with a null axis along rows. Simply reversing light through this same relay does not give a column-resolved readout: reversing a lens that focuses row j onto a point merely re-expands a point source back into a line illuminating the whole of row j; it performs no complementary focusing along columns. A second relay, focusing along rows with a null axis along columns, and sharing the same physical transmittance plane, is required to collect a column-resolved signal. Figure 1 shows both relays schematically. This is additional optical hardware, but of the same kind already present — more lenses, sources, and detectors of an identical class to those already used in the forward path — not a new physical mechanism.

3.3 Closed-form backward operation

Injecting intensity e_j from row j through the second relay and collecting at column detector C_i gives, by the same argument as Eq. (1) applied to the orthogonal relay,

$$C_i = \sum_{j=1}^{M} e_j \, w_{ij}. \tag{3}$$

Setting $e_j = \delta_j$ realises the matrix-transpose product required by Eq. (2), using no component not already justified in Section 2.

4 Signed Error Encoding

δ_j may take either sign, exactly as w_{ij} does. The same differential encoding is reused: $\delta_j = e_j^+ - e_j^-$, $e_j^+ = \max(\delta_j, 0)$, $e_j^- = \max(-\delta_j, 0)$, each driving its own source. Both e_j^+ and e_j^- traverse both

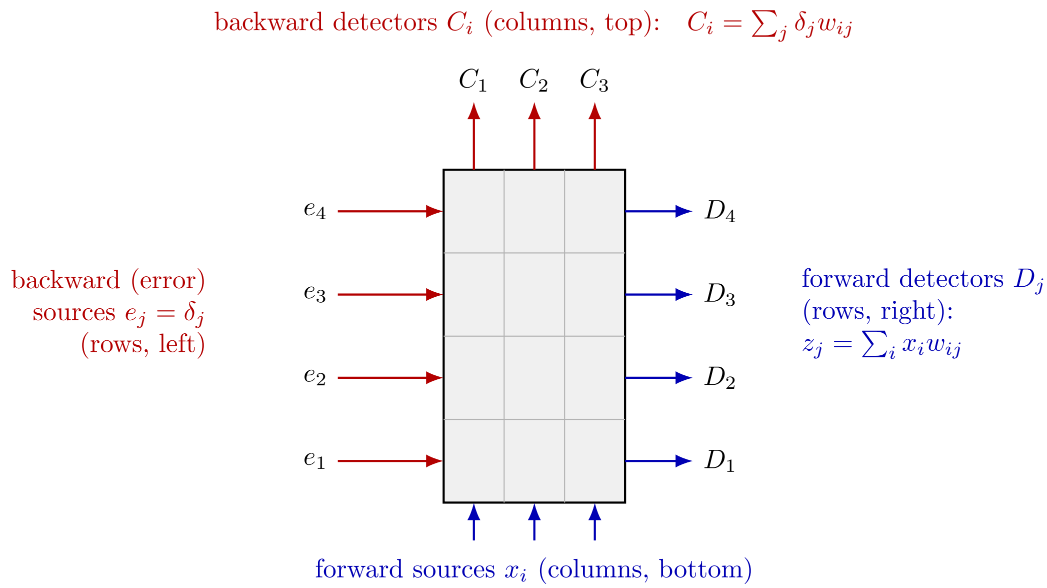

Figure 1: The same physical plane read by two orthogonal relays. Forward (blue): column sources x_i illuminate the plane from below; a relay focusing along columns collects each row onto a dedicated detector D_j, giving $z_j = \sum_i x_i w_{ij}$. Backward (red): row sources carrying the error signal $e_j = \delta_j$ illuminate the plane from the left; a second, orthogonal relay focusing along rows collects each column onto a dedicated detector C_i, giving exactly the matrix-transpose product $C_i = \sum_j \delta_j w_{ij} = (\delta W^\top)_i$, using the direction-independence of passive transmittance and no phase information. The plane itself stores $w_{ij} = w^+_{ij} - w^-_{ij}$ as in the differential push–pull encoding of Section 4.

existing push–pull cells w^+_{ij} and w^-_{ij}, requiring four column-detector banks:

$$\begin{aligned} C_i^{++} &= \textstyle\sum_j e_j^+ w_{ij}^+, & C_i^{+-} &= \textstyle\sum_j e_j^+ w_{ij}^-, \\ C_i^{-+} &= \textstyle\sum_j e_j^- w_{ij}^+, & C_i^{--} &= \textstyle\sum_j e_j^- w_{ij}^-. \end{aligned} \tag{4}$$

A single electronic combination, following directly from expanding $(e^+_j - e^-_j)(w^+_{ij} - w^-_{ij})$ and summing over j, recovers the signed product:

$$\left(\delta W^\top\right)_i = \left(C_i^{++} + C_i^{--}\right) - \left(C_i^{+-} + C_i^{-+}\right). \tag{5}$$

The cost is a doubling of detector banks relative to the unsigned case, the same kind of overhead the forward differential encoding already accepts for signed weights.

5 The Elementwise Gate Is Already Available

The forward measurement that yields z_j also settles, as a single comparator bit, whether $z_j > 0$ — exactly the gate $\mathbb{1}[z_j > 0]$ that Eq. (2) requires. Latching this bit during the forward pass and applying it to $(\delta W^\top)_i$ on the way out of Eq. (5) costs one digital gate per neuron and no additional optics: the rectifying nonlinearity is free in both the forward and the backward direction, for the same underlying reason.

6 The Weight Update: Where the Free Lunch Ends

6.1 A counting argument

The efficiency of both Eq. (1) and Eq. (3) comes from collapsing an $N \times M$ array of products to M (or N) detector readouts by summing along one axis at the point of detection. The weight update $\Delta w_{ij} = -\eta x_i \delta_j$ is itself an $N \times M$ array of independent numbers — there is no legitimate sum to take along either axis, since every cell needs its own value. No arrangement of $N + M$ aggregate detectors can carry $N \times M$ independent numbers; this is a counting argument, not an engineering shortfall, and it holds regardless of optical cleverness.

6.2 A tempting idea, and why it fails

One might let the plane material itself be photosensitive and simply co-illuminate cell (i, j) with the held value x_i from one side and δ_j from the other, letting local exposure dose write Δw_{ij} with no electronics at all. This fails as stated: for *mutually incoherent* beams, absorbed dose is additive, dose $\propto x_i + \delta_j$, not multiplicative — the same incoherence that makes Eq. (1) a sum rather than an interference pattern makes a passive product-writing mechanism unavailable here. We record this explicitly because it is an easy error to wave past rather than rule out.

6.3 Primary proposal: hybrid optical projection, local electronic write

We propose computing Δw_{ij} with a small local electronic multiply-accumulate cell at each (i, j), of the kind already established in resistive/phase-change in-memory crossbar training hardware; no novelty is claimed for this electronic component, only for the optically-derived operands feeding it. Both operands are ordinary electrical signals already present at the architecture's edges: x_i is the same value already used to drive the forward source for column i (latched, not re-measured), and δ_j is the output of Eq. (5) after the gate of Section 5. Distributing these as row- and column-address lines across the plane, exactly as a crossbar distributes write voltages, lets every cell update in parallel, in a time τ_{mac} bounded by the local multiplier's settling time and independent of N and M.

6.4 A speculative extension: two-timescale coherent consolidation

A more ambitious and explicitly more speculative alternative removes the local electronics. The fast, repeated stages — forward inference and the backward projection of Section 3 — remain incoherent, since these execute at high repetition rate and interferometric stability there would be prohibitive. A separate, rare, deliberately slow *consolidation* phase is proposed instead: a holographic write into a photorefractive plane, in which a phase-stabilised replay of the held activations x_i and the projected error δ_j genuinely interferes, so that the resulting index/transmittance change is proportional to the interference cross-term and therefore to $x_i \delta_j$, as in conventional holographic data storage. Because this phase runs once per batch rather than once per inference, its stability budget is amortised over many fast cycles. No precision or signal-to-noise budget is derived for this scheme: doing so honestly would require photorefractive material parameters outside the scope of this paper, and it is listed under open problems (Section 10) rather than under results.

7 A Closed-Form Training Latency Model

A first, naive estimate would simply add a forward pass and a backward pass and call the result done. Two corrections are needed before that estimate can be trusted, one about *how much* optical reading is actually required, and a far more important one about what dominates the budget once a weight is actually changed.

7.1 Exactly how many backward passes are needed

Write the network as L weight planes $W^{(1)}, \dots, W^{(L)}$, with $W^{(l)}$ mapping an N_{l-1}-dimensional input to an N_l-dimensional output (N_0 is the raw input width, N_L the output width). Updating $W^{(l)}$ requires $\delta^{(l)}$ (Eq. (2)), and $\delta^{(L)}$ is available directly and electronically from the loss — no optics needed. For $l < L$, $\delta^{(l)}$ is obtained by the reciprocal-transmittance backward pass of Section 3 *through plane* $W^{(l+1)}$, reading N_l column detectors. Crucially, $W^{(1)}$ itself is *never* read backward: its gradient $\Delta w_{ij}^{(1)} = -\eta\, x_i^{(0)}\, \delta_j^{(1)}$ only needs $\delta^{(1)}$, which already came from the backward pass through $W^{(2)}$, and the already-held input $x^{(0)}$ (dimension N_0, which can be large — 784 for a raw-pixel input — but is never an optical *readout*, only a held forward value). So exactly $L-1$ backward optical passes are needed, through planes $W^{(2)}, \dots, W^{(L)}$, reading $N_1, \dots, N_{L-1}$ detectors respectively:

$$T_{\mathrm{fwd}} = \frac{L}{f_s} + \frac{1}{f_d} \sum_{l=1}^{L} N_l, \tag{6}$$

$$T_{\mathrm{bwd}} = \frac{L-1}{f_s} + \frac{1}{f_d} \sum_{l=1}^{L-1} N_l = \frac{L-1}{f_s} + \frac{1}{f_d}\Big(\textstyle\sum_{l=1}^{L} N_l - N_L \Big). \tag{7}$$

For the representative 784-64-32-10 network ($N_1{=}64, N_2{=}32, N_3{=}10,\ L = 3$), T_{fwd} reproduces PONNIP's own $256\,\mu$s at $f_s{=}20\,$kHz, $f_d{=}1\,$MHz, and $T_{\mathrm{bwd}} = 2/f_s + 96/f_d = 196\,\mu$s. The two stages are of comparable order — the backward pass is in fact slightly *cheaper*, since it both skips one layer's read (the network's largest, N_0, is never read backward) and never needs to read the output layer's own width N_L a second time. We call $T_{\mathrm{floor}} = T_{\mathrm{fwd}} + T_{\mathrm{bwd}} = 452\,\mu$s the *optical floor*: the fastest this architecture's training step could possibly be, if writing the update were free.

7.2 Writing the update is not free

It is not free, and this is the dominant correction. The local electronic multiply-accumulate cell of Section 6 computes Δw_{ij} quickly (at τ_{mac}, microsecond scale), but applying that update means changing the control signal driving the *same physical attenuating cell* used for inference — and that cell's transmittance does not change instantaneously. PONNIP's own latency model (its Eq. 11) already identifies this settling time, T_{settle}, as the slowest timescale in the entire architecture, of order milliseconds for any technology with a mechanical or electro-optic relaxation step, several orders of magnitude slower than the microsecond-scale electronic stages. PONNIP's multi-plane pipeline (its Section 4.2) was specifically designed so that *inference* pays T_{settle} only once, at deployment. Training cannot make that trade: by definition, the weights must keep changing, so T_{settle} is paid again on every update. This is precisely the missing term in a naive forward-plus-backward estimate, and it is the right place to look for where the extra work in training actually goes: not in the optical reads, which are fast and roughly symmetric between the two directions, but in the physical commitment of the new weight value to an analog medium.

Two bounds apply, depending on whether the L planes can settle concurrently (they are physically distinct devices, so nothing in principle forbids it) or only sequentially (if, for instance, a shared control bus limits how many planes can be driven at once):

$$T_{\text{settle,total}} \in \left[T_{\text{settle}},\ L\,T_{\text{settle}}\right]. \tag{8}$$

Per training example, with no batching,

$$T_{\text{train}}(B{=}1)\ \approx\ T_{\text{floor}} + T_{\text{settle,total}}. \tag{9}$$

7.3 Batching amortises the settling cost

The local multiply-accumulate cell of Section 6 can integrate Δw_{ij} electronically over B training examples *before* committing a single write to the optical plane — exactly the role a mini-batch already plays in ordinary stochastic gradient descent, here mapped onto a hardware constraint rather than a statistical one. The optical floor must still be paid once per example (every example needs its own forward and backward pass to produce a gradient), but the settling cost is paid once per *batch*:

$$T_{\text{train}}(B)\ \approx\ T_{\text{floor}} + \frac{T_{\text{settle,total}}}{B}. \tag{10}$$

As $B \to \infty$, $T_{\text{train}}(B) \to T_{\text{floor}}$: training can approach, but on this model never quite reach, the optical floor, since the floor itself already includes a backward pass that pure inference does not need.

8 Numerical Sanity Check

For the representative 784-64-32-10 network at f_s=20 kHz, f_d=1 MHz, Section 7.1 gave $T_{\text{floor}} = 452\,\mu$s. Taking an illustrative $T_{\text{settle}} = 1$ ms — the same order of magnitude PONNIP itself uses qualitatively, not a measured value — Eq. (9) gives, with no batching,

$$T_{\text{train}}(1) \in [\,452\,\mu\text{s} + 1\,\text{ms},\quad 452\,\mu\text{s} + 3\,\text{ms}\,] = [\,1.45\,\text{ms},\ 3.45\,\text{ms}\,] \text{ per example,} \tag{11}$$

i.e. training is genuinely slower than inference under this model — by a factor of roughly 6 to 13 relative to PONNIP's own $256\,\mu$s inference figure — and the gap is entirely attributable to the settling time of the weight plane itself, not to the optical reads. Figure 3 shows how this gap closes as the batch size B grows: at $B = 10$, T_{train} falls to 0.55–0.75 ms; by $B \approx 100$, it is within a few percent of the optical floor. This is a plausibility projection from a simplified model, not a measured result: it neglects settling transients that are not simple step responses, write non-idealities, and any precision loss from finite multiplier resolution.

9 Relation to Existing Approaches

- **In-situ adjoint backpropagation on coherent meshes** [5, 6]: derives the photonic analogue of backpropagation via adjoint variable methods and measures gradients through local interference in a unitary, phase-coherent circuit. This paper targets the structurally different incoherent, attenuation-based class of device, where phase is unavailable by design, and uses the weaker direction-independence of transmittance instead, requiring no interference for the backward projection itself.

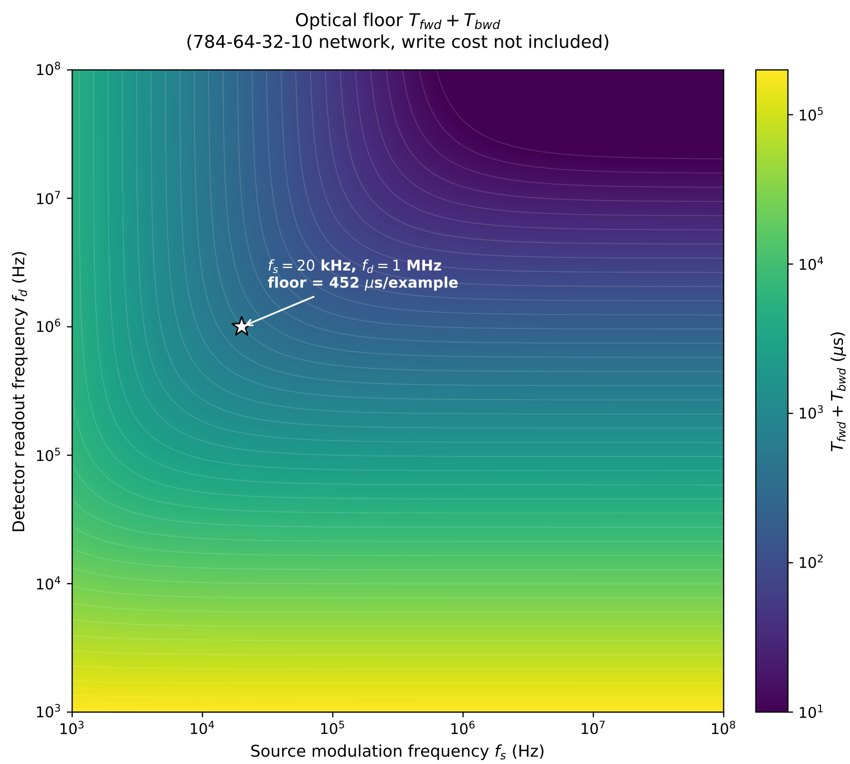

Figure 2: The optical floor $T_{\text{fwd}} + T_{\text{bwd}}$ (Eqs. (6)–(7)) for the 784-64-32-10 network, as a function of source switching frequency f_s and detector readout frequency f_d. This is the fastest the architecture could run if writing the update were free; Figure 3 shows why it generally is not.

- **End-to-end optical backpropagation via saturable absorbers** [7]: resolves the conflicting forward/backward requirements on the nonlinear element by choosing an activation whose physical response is usable in both directions. This paper resolves the same difficulty differently, by observing that the rectifying nonlinearity need not be run backward at all — only its already-latched binary derivative is needed (Section 5).

- **Physics-aware training** [8]: trains a differentiable digital twin and never runs a physical backward pass, a conservative and broadly applicable path across physical substrates. This paper instead asks how far an actual physical backward pass can be pushed for this specific incoherent architecture, as a complementary rather than competing question.

- **Direct feedback alignment** [9], including its photonic implementation on a coherent silicon platform [10]: avoids the transpose-weight requirement altogether by projecting the output error through a fixed random matrix, at the cost of approximate (rather than exact) gradients. This paper keeps exact backpropagation, at the cost of the additional relay hardware of Section 3.

- **Equilibrium propagation** [11]: avoids a separate backward pass altogether via a free/nudged contrastive rule, but is formulated for recurrent, energy-based networks rather than the strictly feedforward, per-layer pipeline assumed here, and is not straightforwardly applicable to this architecture without a substantial reformulation not attempted in this paper.

We have not located, in the literature consulted, a treatment that derives the backward pass for

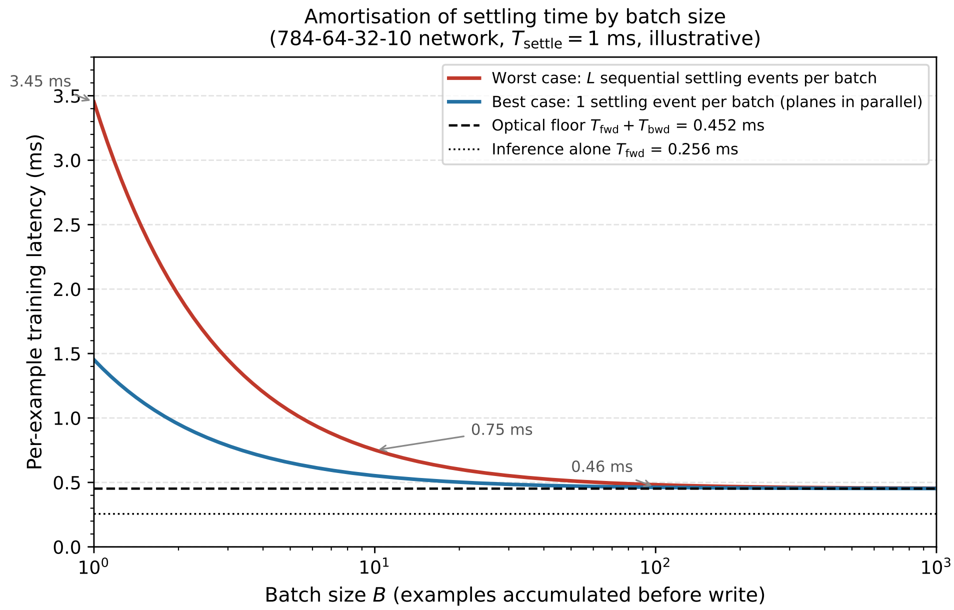

Figure 3: Per-example training latency from Eq. (10) as a function of batch size B, for the 784-64-32-10 network with $T_{\text{settle}} = 1\,\text{ms}$ (illustrative). Without batching, training is dominated by the settling time of the weight plane itself (3–7$\times$ the optical floor); accumulating updates electronically over a batch before committing a single write amortises this cost towards, but never quite down to, the optical floor of Figure 2.

an incoherent attenuation-superposition architecture from transmittance reciprocity alone, in the way Section 3 does; this is stated as the best of our knowledge rather than as a verified claim of priority, which a literature search of this kind cannot establish with certainty.

10 Open Problems

- **Crosstalk and diffraction in the dual relay.** Two orthogonal imaging systems sharing one physical plane have finite resolution and will not perfectly isolate row j from row j' or column i from column i' at realistic component sizes. No such analysis is given here.

- **Precision budget.** No signal-to-noise or effective-bit-width analysis is given for the four-bank recombination of Eq. (5) or for the local multiply-accumulate write. Training requires many repeated small updates rather than a single weight load, and is correspondingly more exposed to analog drift and limited weight resolution than inference is.

- **Material fatigue and dynamic range**, specific to the speculative consolidation scheme of Section 6.4: photorefractive media are known to exhibit saturation and erasure effects under repeated write/erase cycling, not modelled here and assessed as the largest open risk attached to that variant specifically.

- **No energy budget is derived**; a component-level power analysis belongs to a hardware-implementation study, not a theoretical derivation of this kind.

- **Nothing here has been built or simulated end-to-end.** This is a from-first-principles derivation, not an experimental result.

11 Conclusion

This sequel to PONNIP [1] asked whether the same two incoherent primitives — intensity superposition and passive attenuation — that justify PONNIP's simplicity at inference could also supply, on-chip, the gradient information needed to train it. The matrix-transpose product and the elementwise activation gate required by backpropagation can both be obtained this way, without reintroducing the interferometric stability PONNIP was designed to avoid: the transpose product, by exploiting the direction-independence of passive transmittance through a second, orthogonal relay sharing the same physical plane; the gate, by reusing information the forward pass already measures. The weight update itself cannot be obtained this way: a counting argument shows no arrangement of aggregate detectors carries the necessary per-cell information, and a tempting passive photosensitive write is ruled out by the same incoherence that makes the rest of the architecture simple. A conservative hybrid resolution — optically-derived operands feeding a local electronic multiply-accumulate write — is proposed as the primary result, with a more ambitious two-timescale coherent-consolidation scheme flagged as an explicitly open, underived perspective. Once the latency model is required to account for what PONNIP's own analysis already identified as its slowest stage — the settling time of the attenuating plane itself — training is genuinely slower than inference, by roughly one order of magnitude with no batching, falling towards (but never quite reaching) inference-comparable latency as the batch size grows. This is a plausibility projection, not a guarantee, and it leaves several concrete problems honestly unsolved.

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